



**INTRODUCTORY PHYSICS LABORATORY
DEPARTEMEN OF PHYSICS
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SURABAYA**



NAME :
STUDENT ID :
ASSISTANT :
DATE :

EXPERIMENT: SPRING CONSTANT

Objectives :

1. Determining the spring constant using static and dynamic methods.

EXPERIMENT SUMMARY

INTRODUCTION

Springs are among the most fundamental physical systems in mechanics. Their ability to store and release elastic energy makes them a basic model for various oscillatory phenomena in both nature and engineering. The behavior of an ideal spring is described by Hooke's Law, which states that the restoring force produced by a spring is directly proportional to its displacement from the equilibrium position [2]:

$$F = -k\Delta x \quad (1)$$

In that equation, F is the restoring force (N), Δx is the displacement from the equilibrium position (m), and k is the spring constant (N/m) which reflects the stiffness of the spring. The negative sign indicates that the restoring force is always directed opposite to the displacement, causing the system to tend to return to its equilibrium position [5]. In this experiment, the negative sign is not used explicitly because the quantity measured is the magnitude of the force, not its direction. The spring constant, k can be determined using two fundamentally different approaches. The first is the static method, which utilizes the equilibrium condition between the weight of the load, mg and the restoring force of the spring. Under equilibrium conditions, Equation (1) becomes (Giancoli, 2014):

$$k = \frac{mg}{\Delta x} \quad (2)$$

Second, the dynamic method, which is based on the principle of Simple Harmonic Motion (SHM). Serway & Jewett (2012) state that in an ideal spring–mass system, the oscillation period T depends on the total mass of the system m and the spring constant k according to the following equation:

$$T = 2\pi\sqrt{\frac{m}{k}} \quad (3)$$

This equation holds because the angular frequency of the SHM system satisfies $\omega = \sqrt{\frac{k}{m}}$, which, when combined with the relationship $T = \frac{2\pi}{\omega}$ yields Equation (3). By rearranging this equation, the spring constant can be expressed as a function of the mass and the oscillation period [4]:

$$k = \frac{4\pi^2 m}{T^2} \quad (4)$$

It is important to note that the oscillation period in Simple Harmonic Motion (SHM) does not depend on the amplitude, but only on the mass and the spring constant (Giancoli, 2014). This means that the magnitude of the initial displacement does not affect the measured value of T so the dynamic method provides high reliability as long as the displacement used does not exceed the elastic limit of the spring. In the simple harmonic motion equation it shows that the relationship

$\omega = \sqrt{\frac{k_{\text{ef}}}{m}}$ also applies to both series and parallel spring arrangements, where k_{ef} is the effective spring constant of the arrangement. These two methods essentially determine the same physical quantity but through different approaches, which the static method is based on force equilibrium, while the dynamic method is based on the analysis of periodic motion.

METHODOLOGY

A. Tools and Materials

The following are the apparatus and materials used in this practical session:

Tabel. 1 Apparatus and materials for the spring constant experiment.

Apparatus	Quantity	Function
Load	1 buah	Used to provide stiffness to the spring.
Spring	1 buah	Used as the object being observed.
Stopwatch	1 buah	Used to measure the oscillation period.
Stand	1 set	Used to support the spring.
Ruler	1 buah	Used to measure the length of a spring.
Digital Scale	1 buah	Used to measure the mass of a load.

B. Procedure

This practical work is carried out using the following steps:

A. Static Method

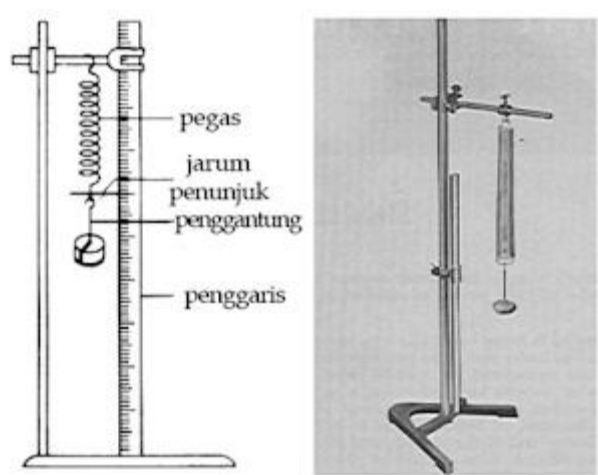


Figure 1. Practicum equipment

1. The first spring was suspended vertically on a stand, then the initial length of the spring (without any load) was measured and set as the zero reference point.
2. Each load was weighed using a digital balance to determine its mass.
3. The first load was hung on the hook at the lower end of the spring, then the total length of the spring was measured, and the extension was calculated by subtracting the spring's initial length. The mass of the load and the resulting extension were recorded.
4. The previous step was repeated by adding loads one at a time until five load variations were obtained. With each additional load, the length of the spring was measured and its extension was recorded.
5. After all the weights have been suspended, they are removed one by one in sequence. For each reduction in load, the length of the spring is measured again, and the difference from its initial length is calculated. The mass of the weight and the resulting difference in length are then recorded.

B. Dynamic Method

1. A load was attached to the lower end of the spring, and the total mass of the load was then determined using a digital balance. Next, the load was displaced by a small distance of 5 cm and released. The time required for four complete oscillations was measured using a stopwatch and recorded.
2. The first step was repeated with two load variations: three weights and all available weights. For each variation, the initial displacement was kept the same. The time required for four complete oscillations was then measured and recorded.

C. Experiment Flowchart

DATA ANALYSIS AND DISCUSSIONS

A. Data Analysis

The following are the tables of measurement result obtained in the laboratory:

Tabel 1. Measurement results using the dynamic method.

Stage	Additional weight (kg)	Δx (cm)	Weight reduction (kg)	Δx (cm)
1				
2				
3				
4				
5				

Tabel 2. Measurement results using the static method.

Repeat to	m_1 (kg)	t (s)	m_2 (kg)	t (s)
1				
2				
3				
4				
5				
Rata-rata				

Approved by:

Laboratory Assitant

B. Data Analysis of Calculation Results

(Calculate the spring constant using both static and dynamic methods)

The following is an example calculation based on measurements taken in the laboratory.

Example calculation:

The following is the table of calculation results obtained:

Table 3. Results of calculations using the static method.

Stage	Additional weight (Kg)	Δx (m)	k (N/m)	Weight reduction (Kg)	Δx (m)	k (N/m)
1						
2						
3						
4						
5						
Average						

Table 4. Results of the calculations using the dynamic method.

Repetition	Period of a single oscillation					
	m_1 (kg)	t (s)	k (N/m)	m_2 (kg)	t (s)	k (N/m)
1						
2						
3						
4						
5						
Average						

C. Graph

(Plot a graph of the change in spring length versus load mass for the static method, and calculate the spring constant based on the graph.)

D. Discussion

CONCLUSION

From this experiment, it can be concluded that:

- 1.

REFERENCES

- [1] M. Abdullah, *Fisika Dasar I*, Edisi Revisi. Bandung, Indonesia: Penerbit ITB, 2016.
- [2] D. C. Giancoli, *Physics: Principles with Applications*, 7th ed. Upper Saddle River, NJ, USA: Pearson Education, 2014.
- [3] D. Halliday, R. Resnick, and K. S. Krane, *Fundamentals of Physics*, 8th ed. Hoboken, NJ, USA: John Wiley & Sons, 2010.
- [4] D. Halliday, R. Resnick, and J. Walker, *Fundamentals of Physics Extended*, 9th ed. Hoboken, NJ, USA: John Wiley & Sons, 2010.
- [5] R. A. Serway and J. W. Jewett, *Physics for Scientists and Engineers with Modern Physics*, 9th ed. Mason, OH, USA: Brooks/Cole, Cengage Learning, 2012.

PRELIMINARY ASSIGNMENT

1. Name and explain three examples of the application of the spring constant in everyday life, along with a brief explanation of the role of the spring constant in each of these applications.
2. What is the fundamental difference between the static and dynamic methods for measuring the spring constant?. Explain the advantages and disadvantages of each method.
3. Derive the equation for the spring constant in the dynamic method $\left(k = \frac{4\pi^2 m}{T^2} \right)$ from the relationship of Simple Harmonic Motion (SHM).